

# Semantic Tags & HTML Layout

Coding and Web Development | Lecture 07 | By - Nishant Sir | Summary Notes

## 1 Semantic Tags

**Semantic Tag** = ■■■ HTML tag jiska **naam khud uska meaning/content bata deta hai** — sirf tag ka naam dekh kar pata chal jata hai andar kis tarah ka content hoga.

Non-semantic tag (jaise `<div>`, `<span>`) sirf ek generic container hote hain — inke andar kya content hai, ye samajhne ke liye humein content ko khud padhna padta hai / ya class/id naam se guess karna padta hai (keyword ke through).

### Non-Semantic Tags (No fixed meaning)

- `<div>` → generic block container
- `<span>` → generic inline container

### Semantic Tags (Meaning is clear)

- `<table>` → tabular data
- `<form>` → user input form
- `<p>` → paragraph text
- `<strong>` → important / bold text

*Note: `<p>` tag ka naam khud batata hai ki andar paragraph content hoga — isi liye ye semantic hai.*

## 2 Website Layout (Blueprint)

Kisi bhi website ka basic structure/blueprint semantic layout tags se banaya jata hai, taaki page ka structure clear ho (jaise Flipkart website: Header, Sidebar/Filters, Main content, Footer).

| Tag                          | Kaam (Purpose)                                                                                         |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>&lt;header&gt;</code>  | Page ka top section — logo, title, search bar                                                          |
| <code>&lt;nav&gt;</code>     | Navigation menu / navbar (nav = navigation ka short form)                                              |
| <code>&lt;main&gt;</code>    | Page ka main / central content                                                                         |
| <code>&lt;article&gt;</code> | Independent, self-contained content — jaise blog post, newspaper article (e.g. "How to learn Backend") |
| <code>&lt;section&gt;</code> | Page ka ek logical/grouped part (ek page ke andar multiple sections)                                   |
| <code>&lt;aside&gt;</code>   | Sidebar — main content se related extra info (jaise Filters sidebar)                                   |
| <code>&lt;footer&gt;</code>  | Page ka bottom section — copyright, links, contact info                                                |

**Nesting example:** `<section>` aur `<aside>` generally `<article>` ke andar bhi use ho sakte hain (article → section, article → aside) — page ke structure ko logically group karne ke liye.

*Semantic layout tags → SEO aur accessibility (screen readers) dono ke liye better hote hain, kyunki structure clearly define hota hai.*

## 3 Media Tags

### Image Tag

- **src** → Source — image ka link / path
- **alt** → Alternative Text — image load na ho to ye text dikhta hai (accessibility ke liye bhi useful)
- **height & width** → image ka size define karne ke attributes

### Figure & Figcaption

Caption text

<figure> ke andar image ke sath uska caption bhi semantically group ho jata hai.

### Video Tag

- **controls** attribute → play/pause, volume jaise video controls dikhata hai
- Video ka **thumbnail** bhi set kiya ja sakta hai (poster attribute se)

### Audio Tag

<audio> tag video tag ki tarah hi kaam karta hai (src + controls attribute), bas isme sirf audio play hota hai, video nahi.

### Iframe Tag

<iframe> ka use tab hota hai jab kisi **dusri website ka content** apni website ke andar embed/show karna ho (jaise YouTube video ya Google Map embed karna).

## 4 HTML Comments

Comments code ke andar likhe jaate hain sirf **developer ki samajh** ke liye — browser inhe render nahi karta / ignore kar deta hai.

- Shortcut: **Ctrl + /** se line ko comment/uncomment kar sakte hain

## 5 HTML Entities

HTML mein kuch characters **reserved** hote hain (jaise <, >, &) — inhe seedha likhne par browser confuse ho sakta hai ki ye HTML tag hai ya normal text. Isliye special **Entity Codes** use kiye jaate hain in special characters ko show karne ke liye.

| Character | Entity Code           | Meaning      |
|-----------|-----------------------|--------------|
| <         | <code>&amp;lt;</code> | Less than    |
| >         | <code>&amp;gt;</code> | Greater than |

|       |                                |                    |
|-------|--------------------------------|--------------------|
| &     | <code>&amp;amp ; amp ;</code>  | Ampersand          |
| "     | <code>&amp;amp ; quot ;</code> | Double quote       |
| Space | <code>&amp;amp ; nbsp ;</code> | Non-breaking space |

---

*Notes prepared from Lecture 07 handwritten notes.*